

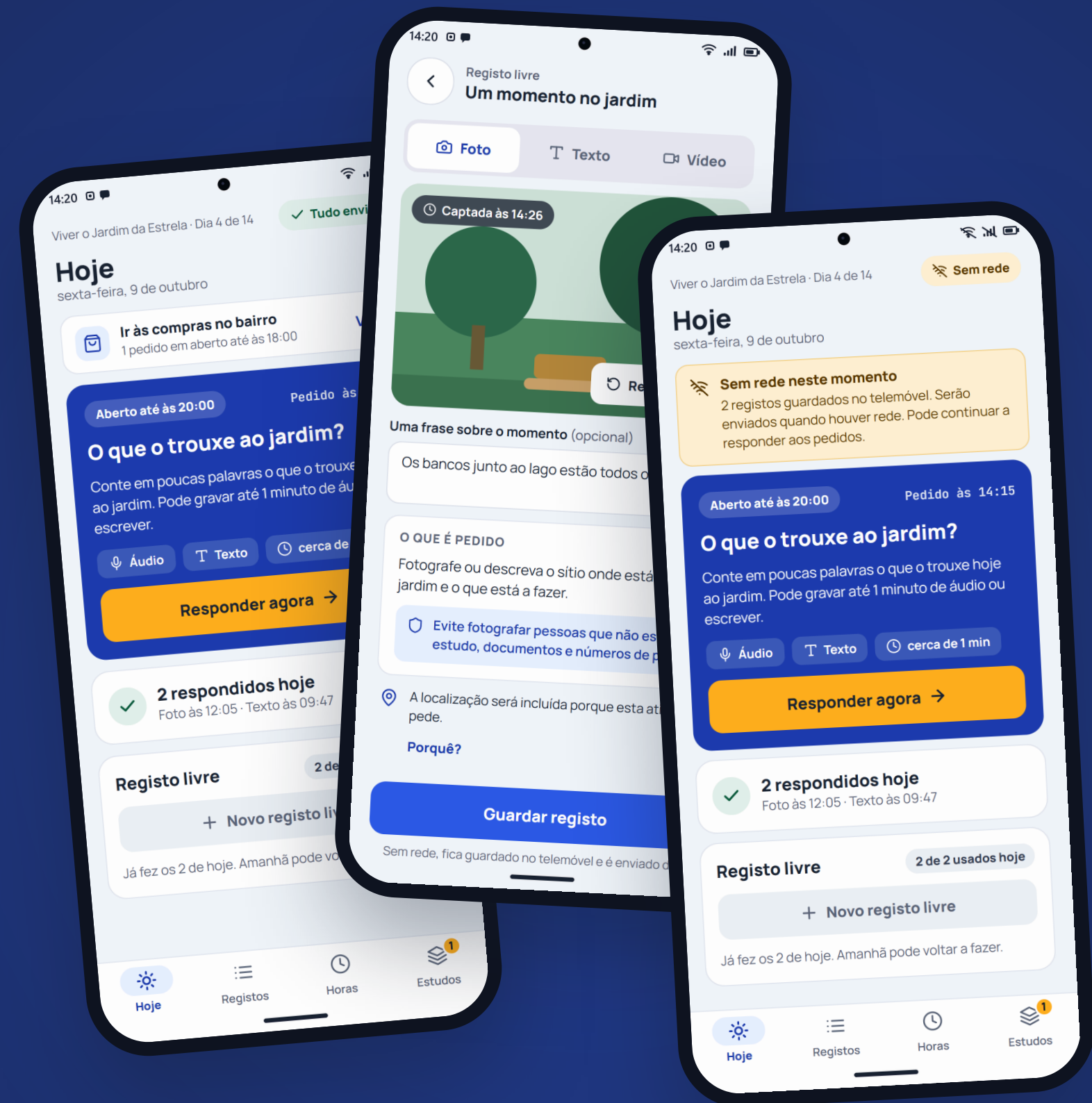


RESEARCH DIARIES ON THE PHONE

Record life as it happens

In paper diaries, participants reported 90% adherence and actually achieved 11%.

11% actual adherence **versus** **90%** reported adherence



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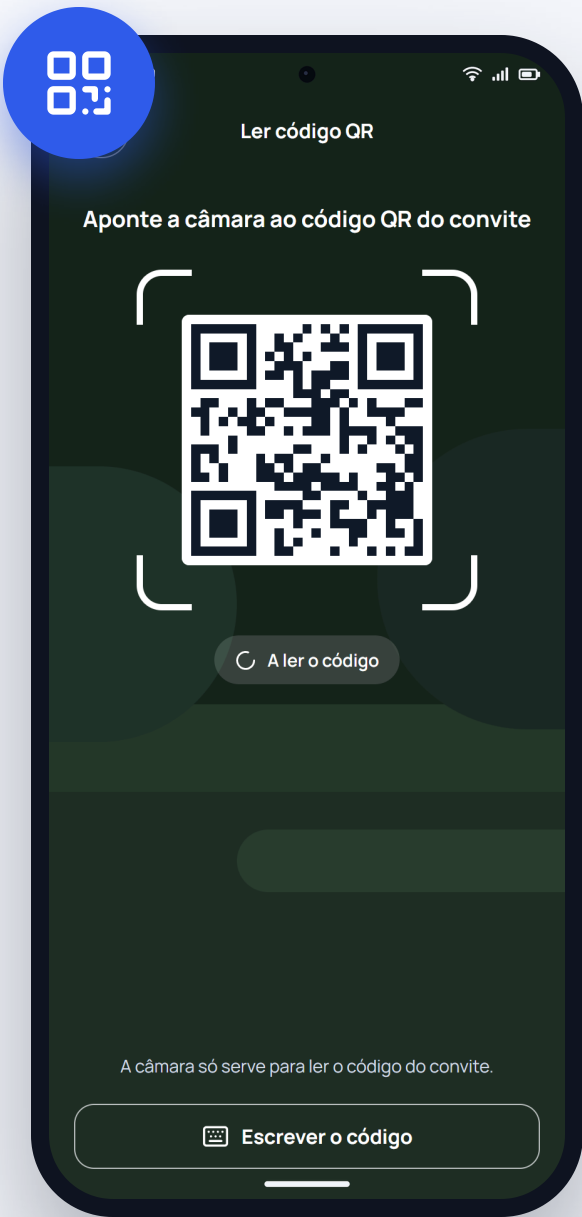
BSc in Computer Engineering, 5th semester · IADE, Universidade Europeia · 2026-2027

Joins with a code and records on the spot

Interface in Portuguese

Optional account

Deletes data in the app



Joins by QR or code

Or by an invitation from the volunteer pool.



Sets their hours

Prompts only arrive within them.



Receives prompts

A notification says when to record.



Records in any format

Text, photo, video, audio or a scale from 1 to 7.



Offline, nothing lost

Stays on the phone and is sent without duplicates.

Designs the study and follows it in real time

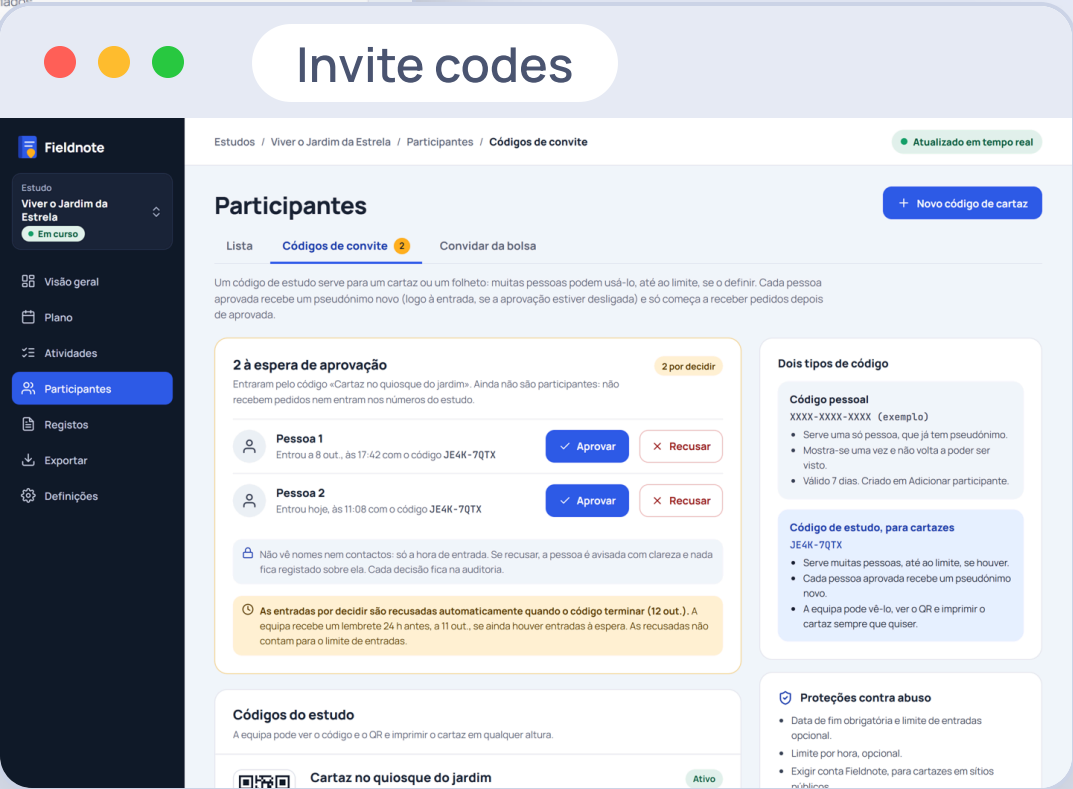
 **Creates studies and activities**
Sets when to ask for each one.

 **Invites in three ways**
Personal invite code, poster code or volunteer pool.

 **Follows adherence**
Entries and responses in real time.

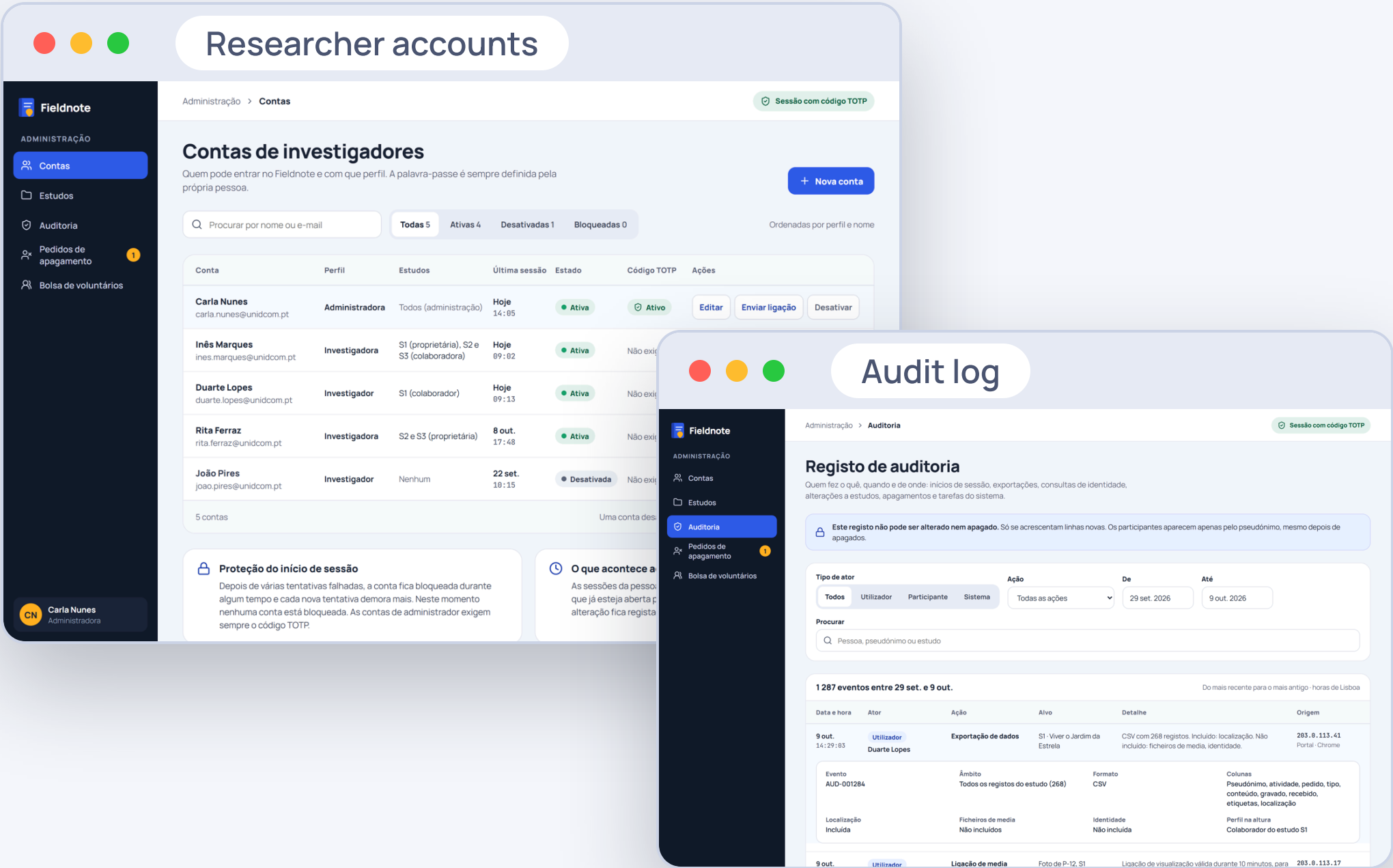
 **Sees when it was recorded**
Recorded time and received time of each entry.

 **Tags and exports**
No location or media by default.



Accounts, pool and audit in one place

- Accounts with TOTP code
- All studies
- Volunteer pool
- Audit
- Deletion requests



PRIVACY AND SECURITY

Participant data stays protected

Pseudonym
The participant appears as P-07.

Authentication and roles
Each role sees only what is theirs.

Media by signed URL
Short-lived links.

No location in photos
EXIF data is removed.

An audit log nobody edits
Who did what, and when.

Deleting means deleting
Hidden now, deleted on day 7.

Three nodes, no entry lost

Stack validated by the PDS and SD lecturers

Architecture

One Docker Compose host

Tailscale private network · 2 Traefik gateways

Node 1

Node 2

Node 3

API
one replica per node

PostgreSQL 17
Patroni + etcd

RabbitMQ
quorum queues

Garage (S3)
3 copies per file

Stated limitation: the three nodes share one physical machine.

Fault tolerance

If any node goes down, it keeps receiving entries. A confirmed entry is never lost.

Demonstration by fault injection

- containers
- whole node
- DB primary
- storage node
- isolated network

Scheduler (AI)

Deciding when to ask for each activity is a constraint satisfaction problem (CSP).

- availability
- activity window
- minimum gap
- daily limit
- max. per 15 min

Backtracking, MRV, LCV, forward checking and AC-3.

APP	PORTAL	SERVER	DATA	INFRA
React Native · Expo TypeScript · SQLite	React · Vite	Python 3.12 · FastAPI ffmpeg · Expo Push	PostgreSQL 17 · etcd RabbitMQ · Garage	Docker Compose · Tailscale Traefik · Prometheus · Grafana



Questions?

 github.com/Cabacadosinformaticos/fieldnote

9 Oct pitch
11 Oct 1st delivery
13 Nov prototype
18 Dec final delivery

Software Development Project

Software Engineering

Distributed Systems

Artificial Intelligence

Information Security

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